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6. Simulation of Handoff Scenario in a Mobile Network Using Figma

Aim:

To design a visual simulation in Figma illustrating the handoff process in a mobile network when a mobile user moves from one cell coverage area to another.

Procedure:

1. Open Figma and create a new design file.
2. Draw two cell towers representing Base Station 1 (BS1) and Base Station 2 (BS2).
3. Draw circular coverage areas around both base stations.
4. Add a mobile phone icon representing the moving user.
5. Use arrows to show the movement of the mobile user from Cell 1 to Cell 2.
6. Create frames illustrating the stages: initial connection, movement toward the cell boundary, handoff initiation, and connection to the new base station.
7. Use different colors to represent the active and inactive connections.
8. Connect the frames using prototype interactions to demonstrate the handoff sequence.

01 - Handoff Introduction

MOBILE COMPUTING 01 / 04

Mobile Network Handoff

Maintaining connectivity while moving.

A CONNECTED JOURNEY



WHAT IS HANDOFF?

Handoff transfers an active mobile connection from one serving cell to another as the user moves.

FOLLOW THE CONNECTION

- Connect
- Compare
- Transfer

One active call. Two cells. A seamless transition.

Start Simulation →

02 - Initial Connection

MOBILE COMPUTING 02 / 04

Step 1: Initial Connection

The call begins in Cell A.

CONNECTED TO A



CONNECTION STATUS

Serving Cell:	Cell Tower A
Signal Strength:	Strong
Call Status:	Active
Connection:	Established

WHAT HAPPENS

The mobile device communicates through its serving base station.

Previous

Next: Monitor Signal →

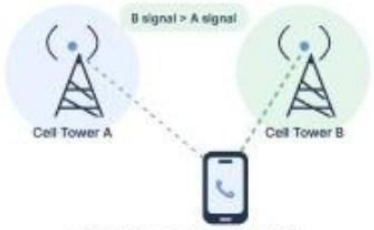
03 - Handoff Decision

MOBILE COMPUTING 03 / 04

Step 2: Handoff Decision

A neighboring cell offers a better signal.

COMPARING SIGNALS



Dashed lines - Serving A / Candidate B

CONNECTION STATUS

Serving Cell:	Cell Tower A
Signal from A:	Weakening
Signal from B:	Stronger
Call Status:	Active

THE HANDOFF DECISION

The network decides to transfer the connection when a suitable neighboring cell offers a stronger signal.

Previous | Next: Handoff →

04 - Handoff Complete

MOBILE COMPUTING 04 / 04

Step 3: Handoff Complete

New serving cell. Same active call.

CONNECTED TO B



Solid line - Active connection

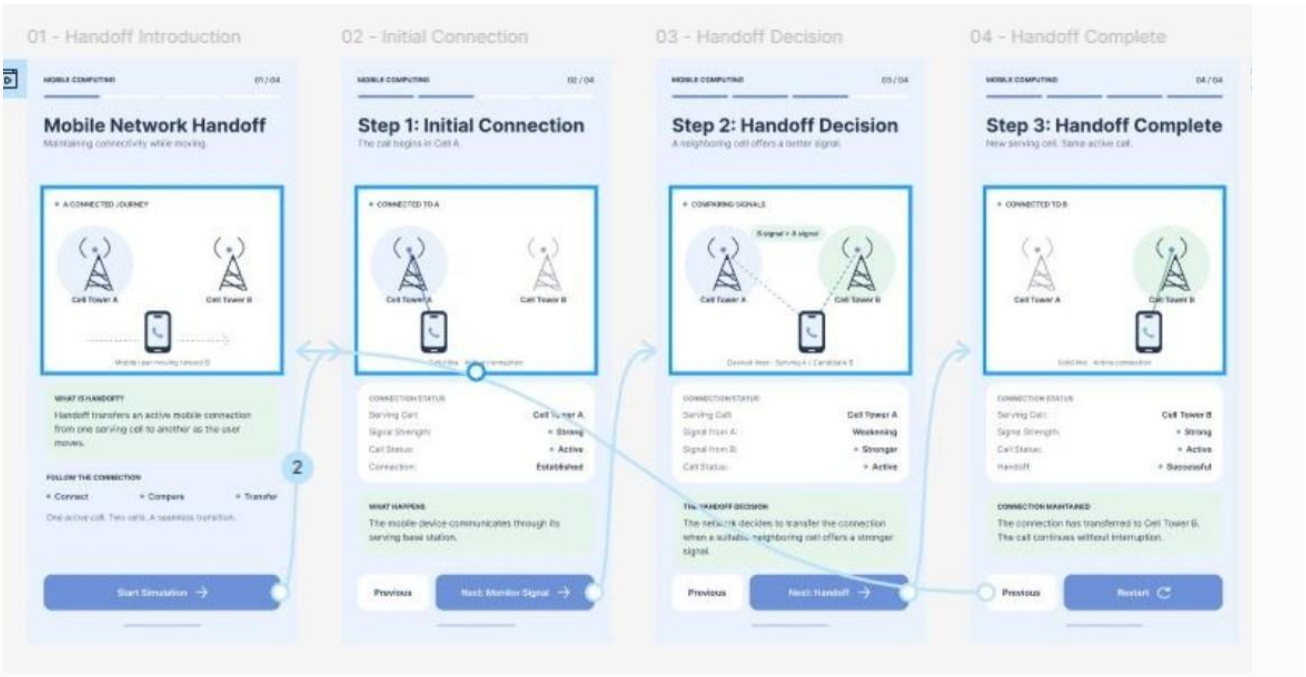
CONNECTION STATUS

Serving Cell:	Cell Tower B
Signal Strength:	Strong
Call Status:	Active
Handoff:	Successful

CONNECTION MAINTAINED

The connection has transferred to Cell Tower B. The call continues without interruption.

Previous | Restart ↻



Result:

A Figma prototype illustrating the mobile network handoff process was designed, showing how an ongoing connection is transferred from one base station to another without interrupting the communication.